

VerbaTask

Autonomous Voice-First WhatsApp ERP & Conversational CRM for
Micro-Merchants

Installation & Evaluator Manual

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Track: Small & Mid-Sized Business (SMB) Retail Technology

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1. Project Overview & System Requirements

This manual covers installation, configuration and a guided evaluation walkthrough for VerbaTask. Follow section 2 carefully before testing, it explains a Meta platform restriction that, if skipped, will make the bot appear completely non-functional to an evaluator even though the system is working correctly.

1.1 Required Software

Requirement	Minimum version
Node.js	v20 or later
MongoDB	v6 or later (local instance or MongoDB Atlas)
npm	bundled with Node.js
git	any recent version

1.2 Required API Keys & Credentials

Credential	Obtained from
Groq Cloud API Key	console.groq.com, used for Whisper Large-v3 speech-to-text
Meta WhatsApp Cloud API credentials	developers.facebook.com, phone number ID, access token, app secret
Alibaba Cloud DashScope API Key	Alibaba Cloud Model Studio console, used for Qwen-2.5
Gemini API Key and/or ElevenLabs API Key	Used by the tiered text-to-speech pipeline (Edge TTS requires no key and serves as the no-cost fallback tier)

2. Critical Notice for Evaluators & Testers

⚠ READ BEFORE TESTING, Meta WhatsApp Cloud API Sandbox Restriction

This project uses the Meta WhatsApp Cloud API in its free Development Mode / test tier. In this mode, the WhatsApp Business number will not reply to messages from an arbitrary phone number, only numbers that have been explicitly added and verified as "testers" in the Meta developer dashboard can send or receive messages. This is a platform-level restriction, not a defect in VerbaTask. If an evaluator messages the test number from an unregistered phone, no reply will arrive and the system may incorrectly appear broken.

2.1 Step-by-Step: Adding an Evaluator's Phone Number

1. Log into developers.facebook.com and select the VerbaTask Meta App.
2. In the left sidebar, navigate to WhatsApp → API Setup.
3. Scroll to the section labeled "Step 5: Select phone numbers."
4. Click the "To" dropdown and choose "Manage phone number list."
5. Add the evaluator's phone number with the full country code (for example, +92 300 1234567).
6. The evaluator will receive a WhatsApp message containing a 6-digit verification code, enter that code in the dashboard to confirm the number.
7. Once verified, that number can send and receive voice notes, text messages and interactive replies with the VerbaTask test business number.

This process takes under two minutes per evaluator and only needs to be done once per phone number, per Meta app.

3. Backend Installation & Configuration

3.1 Clone and Navigate

```
git clone https://github.com/soban-iftikhar/VerbaTask>
cd VerbaTask/backend
```

3.2 Environment Configuration

Create a .env file in the backend/ directory by following .env.example

3.3 Install Dependencies

```
npm install
```

3.4 Run the Automated Test Suite

Confirm the backend is in a known-good state before proceeding, all 39 tests across 6 suites should pass with zero failures:

```
npm test
```

3.5 Start the Development Server

```
npm run dev
```

3.6 Expose the Webhook and Register It with Meta

WhatsApp requires a public HTTPS URL to deliver webhook events. During development, expose the local server with a tunneling tool:

```
ngrok http 8080
```

Copy the generated HTTPS URL, then in the Meta dashboard go to WhatsApp → Configuration → Webhook and set the Callback URL to:

```
<your-ngrok-url>/webhook/whatsapp
```

Set the Verify Token field to match WEBHOOK_VERIFY_TOKEN from your .env file exactly, click "Verify and save," then subscribe to the "messages" field. Confirm the app is linked to the WhatsApp Business Account by checking the subscribed_apps endpoint via Graph API Explorer, a webhook that is configured but not subscribed at the account level will appear correctly set up while silently never receiving real messages.

4. Frontend Installation & Setup

```
cd ../frontend npm install
```

Configure the API base URL in frontend/.env.local (or the equivalent Vite env file):

```
VITE_API_BASE_URL=http://localhost:8080
```

```
npm run dev
```

The dashboard will be available at the local Vite dev server address (typically http://localhost:5173) and will connect to the backend's Socket.IO channel for live updates as WhatsApp activity occurs.

5. Evaluator Walkthrough & Test Script

Complete section 2 before beginning, the evaluator's phone number must be added and verified in the Meta dashboard, or none of the scenarios below will produce a reply.

Scenario A, Voice-First Urdu Sale

Send a voice note to the VerbaTask test number saying (in Urdu): "اول نقد □ دو کلو" (two kilograms of rice, cash). Expect a spoken Urdu voice note in reply confirming the sale, followed by a short text receipt. This demonstrates the full voice-in, voice-out pipeline described in section 4.2 of the System Design Report.

Scenario B, Cross-Lingual & Misspelled Sale

With "Rice" already in the merchant's inventory, send either a voice note saying "2 chawal sadapay" or a typed message reading "2 riece cash." In both cases, verify the system correctly resolves the item to "Rice" rather than returning an "item not found" error, this demonstrates the cross-lingual matching cascade described in section 4.3.

Scenario C, WhatsApp Guided Order

Type "order." Tap an item from the presented list. Reply with a quantity when prompted. Confirm that the payment-method buttons shown reflect only the payment channels this specific test merchant has configured (see Scenario D), not a fixed generic list.

Scenario D, Store Settings & Profiling

Using an authenticated request (or the dashboard UI, once implemented), call GET and PATCH on /api/merchant/profile to rename the store and update business details and GET and PUT on /api/merchant/payment-methods to change which payment channels are active. Repeat Scenario C afterward to confirm the guided-order payment buttons update accordingly.

6. Troubleshooting & Common Pitfalls

Symptom	Likely cause	Fix
No reply to any message, from any number	App not subscribed to the WhatsApp Business Account at the account level	Verify and re-POST the subscription via Graph API Explorer's <code>subscribed_apps</code> endpoint (see section 3.6).
No reply, but only from the evaluator's specific number	Evaluator's number not added/verified as a tester	Follow the step-by-step process in section 2.1.
Webhook returns 500 on every request	WHATSAPP_APP_SECRET missing or incorrect in .env	Copy the App Secret exactly from Meta App Settings → Basic.
"Invalid API-key" from Qwen/DashScope calls	Wrong DashScope region endpoint for the account (mainland vs. international)	Confirm whether the API key was issued on the international or mainland console and match the corresponding base URL.
Voice replies fail or are silent	All TTS tiers exhausted or misconfigured	Confirm at least one of GEMINI_API_KEY / ELEVENLABS_API_KEY is set; Edge TTS requires no key and should always succeed as the final fallback tier.
ngrok URL stops working mid-session	Free-tier ngrok issues a new URL on every restart	Re-copy the current ngrok URL into the Meta webhook Callback URL field and re-verify.